Case 03 · Architecture chapter: Removal Sample Audit

Following chapter 3 (architecture) of the Magshimim project-book template. Characters are fictional.

Introduction

The spec described freeze, draw, strata, app, assignment, estimate. The builder's question: which programs exist, where each one sits, and what the system must receive from outside.

Plots

Plot	What runs there
Audit unit	Export, freezer, sampler, assignment, findings store, estimator, report, repro pack
Field	Field app and sync
Public	Report and reproduction pack
Checker pool	Independent checkers. A plot with no owner.

Components

Component	Role
ECI export	The district's deletion list from the Commission
Freezer (SHA-256)	Locks the list and publishes its hash
NIST Beacon	Public seed
Sampler (Python)	Reproducible stratified draw
Assignment (PostgreSQL)	Assigns each case to a checker
Verifier pool	Independent checkers; **outside the boundary**
Field app (PWA, WebAuthn, IndexedDB)	Signed findings, offline
Sync (HTTPS)	Uploads findings when coverage returns
Findings store (PostgreSQL)	Findings without names, two years
Estimator (SciPy)	Corrected rate and interval
Report (HTML)	District map and threshold
Repro pack (ZIP)	Seed, frozen list, results

Decisions

- **Sample or full count?** Sample.
- **Public seed?** Yes.
- **Closed door?** Three visits.
- **Who knocks on the door?** An independent pool, and nobody to establish it.

Build order

Sprint	Goal
1	Freeze and draw
2	Estimate and pack
3	Field app
4	Assignment
5	Report

Closing check

Everything inside the boundary works. But the answer "does this person live here?" needs an independent checker, and the checker pool is a plot with no owner: no body establishes it. Those actually available are

the booth-level officers of the same machinery that deleted. The POC already showed what then happens to the estimate. **Verdict: the circle does not close. Falls at architecture.**